



Abdul *Sattar*
Mushtaque *Ali*
Ibrar *Ali*

FINAL PROJECT PROPOSAL

JAVA APPLICATION

SUKKUR IBA UNIVERSITY



ABOUT

Project Title: ChatCodeX

Group Name: Code Crusaders

Course Name: Data Structures

Instructor's Name: Sir Abbas Mehdi

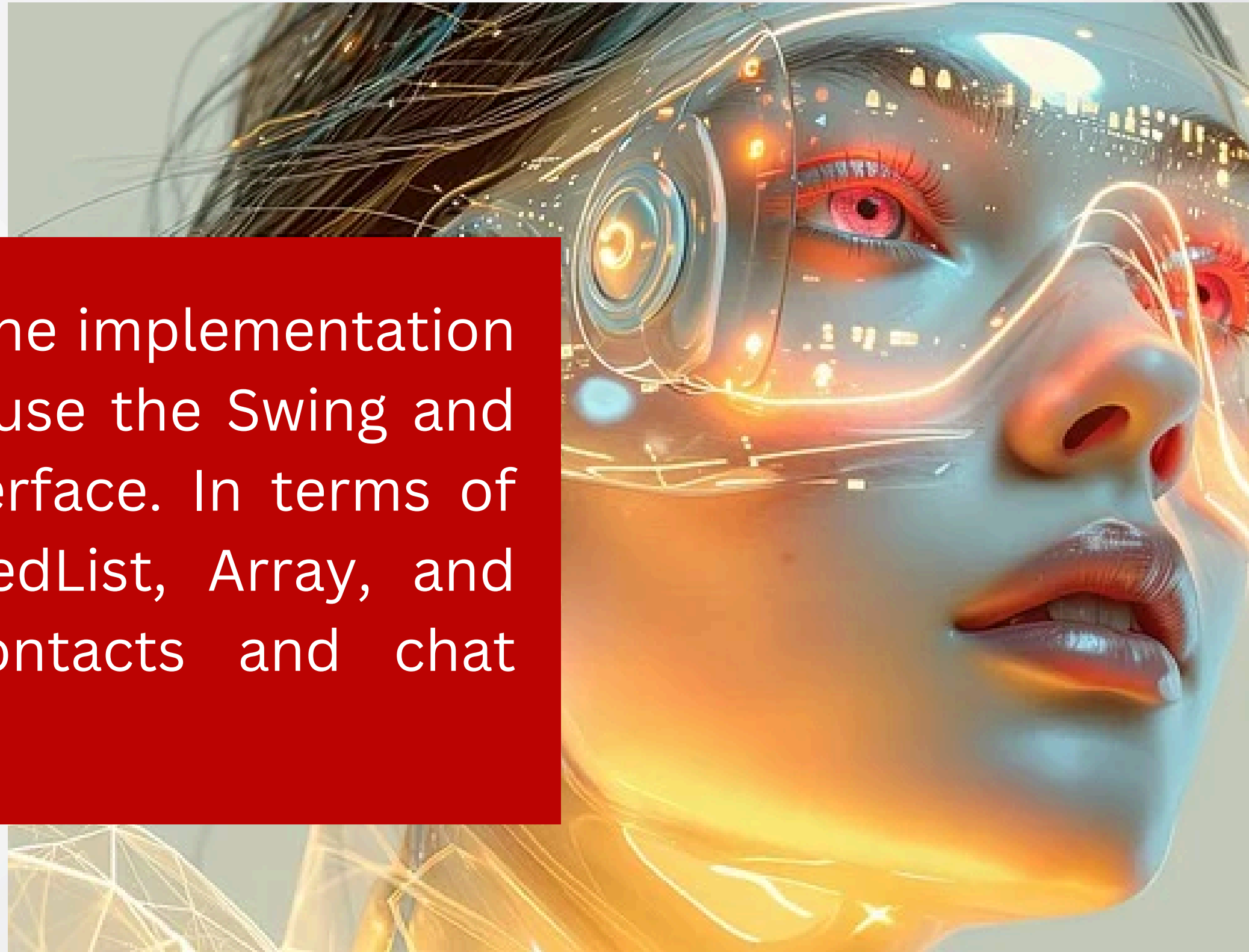
GroupMembers:

Ibrar Ali, Mushtaque Ali & Abdul Sattar



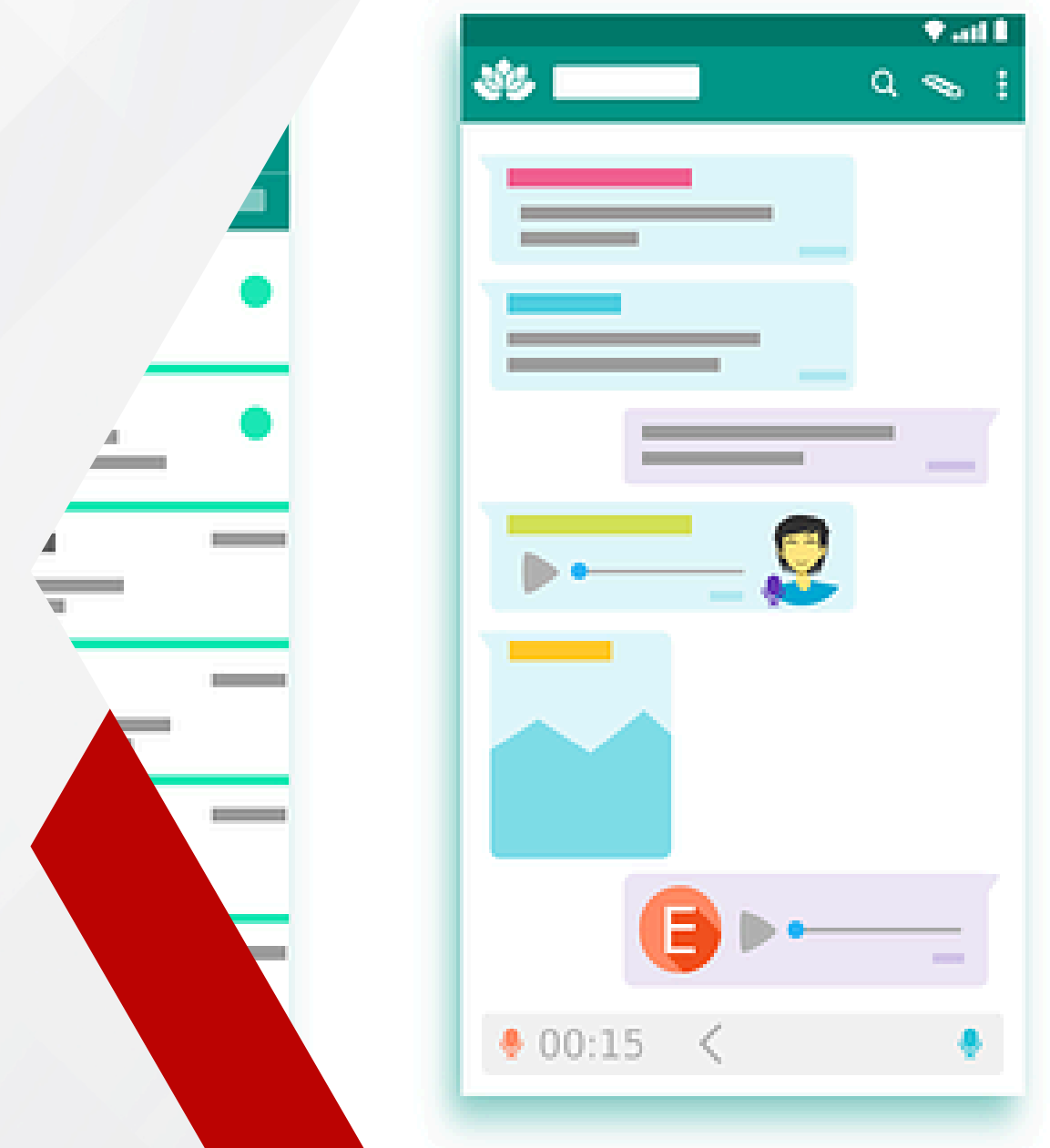
INTRODUCTION

Our project is based on a GUI with the implementation of data structures in Java. We will use the Swing and AWT libraries for the Java GUI interface. In terms of data structures, we will use LinkedList, Array, and Queue to manage and store contacts and chat messages efficiently.



OBJECTIVES

1. Develop a messaging application with two mobile screens (Person A and Person B).
2. Implement functionalities to message each other, save new contacts, and search and delete contacts with chat.
3. Ensure the application uses efficient data structures for managing contacts and messages.



METHODOLOGY

Data Structures Used

- LinkedList
- Array
- Queue

Libraries

- Swing (Java)
- AWT (Java)
- java.util (Queue, LinkedList)





IMPLEMENTATION

The project will be implemented using Java. The user interface will be developed using the Swing and AWT libraries. Contacts will be managed using LinkedList, while chat messages will be handled using Queues. The application will be tested thoroughly to ensure all functionalities work as intended.

FUNCTIONALITY

When the user clicks on the logo, the mobile screen will open, displaying an empty screen if no contacts exist. When the user clicks on "new contacts," a dialog box will appear, prompting for details like name and number. Adding a contact will automatically create a new chatbox, enabling communication with that contact. When the user goes into chat and types a message, the receiver's screen will show a popup, allowing them to reply. Both sides of the conversation will be displayed. Users can exit chats using the back button and delete contacts and chats using the options available via the three-dot button.



TECHNICAL DETAILS

Contacts will be saved in a LinkedList.

- Users will be able to delete contacts,
- update contact names, and search for contacts.

Chat messages will be saved in Queues:

- Sender A's messages in Queue1
- Receiver/Person B's messages in Queue2



RESOURCES



**VISUAL STUDIO
CODE**

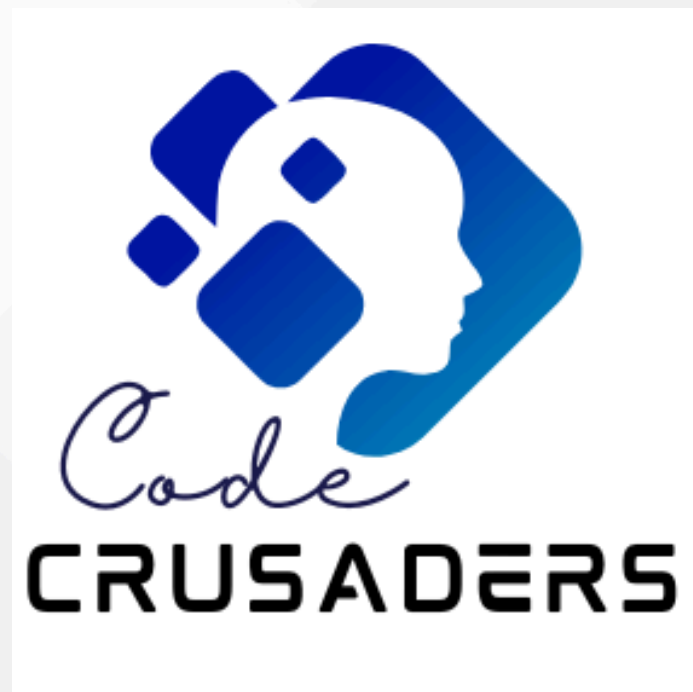
eclipse



ECLIPSE IDE



**MICROSOFT
COPILOT**



THANK YOU

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